Summary

4 annotations on 3 pages

x4

- **Complex instruction set computer (CISC)**
  - Many different instructions with many different formats
    - But, only small subset encountered with Linux programs

#1 *p. 3*

- **x86-64**
  - The standard
  - `shark> gcc hello.c`
  - `shark> gcc -m64 hello.c`

#2 *p. 10*

- **No aggregate types such as arrays or structures**
  - Just contiguously allocated bytes in memory

#3 *p. 15*

Machine-Level Programming I: Basics

15-213/18-213: Introduction to Computer Systems
5th Lecture, September 13, 2016

Instructor:

Phil Gibbons

Today: Machine Programming I: Basics

- **History of Intel processors and architectures**
- **Assembly Basics: Registers, operands, move**
- **Arithmetic & logical operations**
- **C, assembly, machine code**

Intel x86 Processors

- **Dominate laptop/desktop/server market**

- **Evolutionary design**
  - Backwards compatible up until 8086, introduced in 1978
  - Added more features as time goes on

- **Complex instruction set computer (CISC)**
  - Many different instructions with many different formats
    - But, only small subset encountered with Linux programs
  - Hard to match performance of Reduced Instruction Set Computers (RISC)
  - But, Intel has done just that!
    - In terms of speed. Less so for low power.

Intel x86 Evolution: Milestones

Name	*Date*	*Transistors*	*MHz*
8086	**1978**	**29K**	**5-10**

- First 16-bit Intel processor. Basis for IBM PC & DOS
- 1MB address space

386	**1985**	**275K**	**16-33**

- First 32 bit Intel processor , referred to as IA32
- Added “flat addressing”, capable of running Unix

Pentium 4E	**2004**	**125M**	**2800-3800**

- First 64-bit Intel x86 processor, referred to as x86-64

Core 2	**2006**	**291M**	**1060-3333**

- First multi-core Intel processor

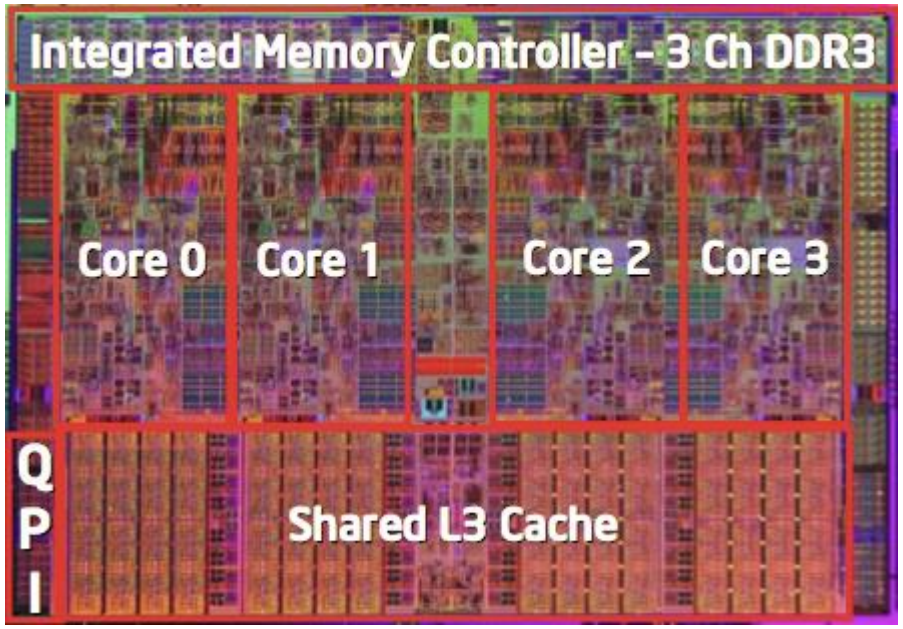
Core i7	**2008**	**731M**	**1600-4400**

- Four cores (our shark machines)

Intel x86 Processors, cont.

Machine Evolution

386	1985	0.3M
Pentium	1993	3.1M
Pentium/MMX	1997	4.5M
PentiumPro	1995	6.5M
Pentium III	1999	8.2M
Pentium 4	2000	42M
Core 2 Duo	2006	291M
Core i7	2008	731M



Added Features

- Instructions to support multimedia operations
- Instructions to enable more efficient conditional operations
- Transition from 32 bits to 64 bits
- More cores

Intel x86 Processors, cont.

- **Past Generations**

		Process technology
1st Pentium Pro	1995	600 nm
1st Pentium III	1999	250 nm
1st Pentium 4	2000	180 nm
1st Core 2 Duo	2006	65 nm

- **Recent Generations**

1.	Nehalem	2008	45 nm
2.	Sandy Bridge	2011	32 nm
3.	Ivy Bridge	2012	22 nm
4.	Haswell	2013	22 nm
5.	Broadwell	2014	14 nm
6.	Skylake	2015	14 nm

- **Upcoming Generations**

Kaby Lake	2016?	14 nm
Cannonlake	2017?	10 nm

Process technology dimension = width of narrowest wires (10 nm ≈ 100 atoms wide)

2016 State of the Art: Skylake

- **Mobile Model: Core i7**
  - 2.6-2.9 GHz
  - 45 W
- **Desktop Model: Core i7**
  - Integrated graphics
  - 2.8-4.0 GHz
  - 35-91 W
- **Server Model: Xeon**
  - Integrated graphics
  - Multi-socket enabled
  - 2-3.7 GHz
  - 25-80 W

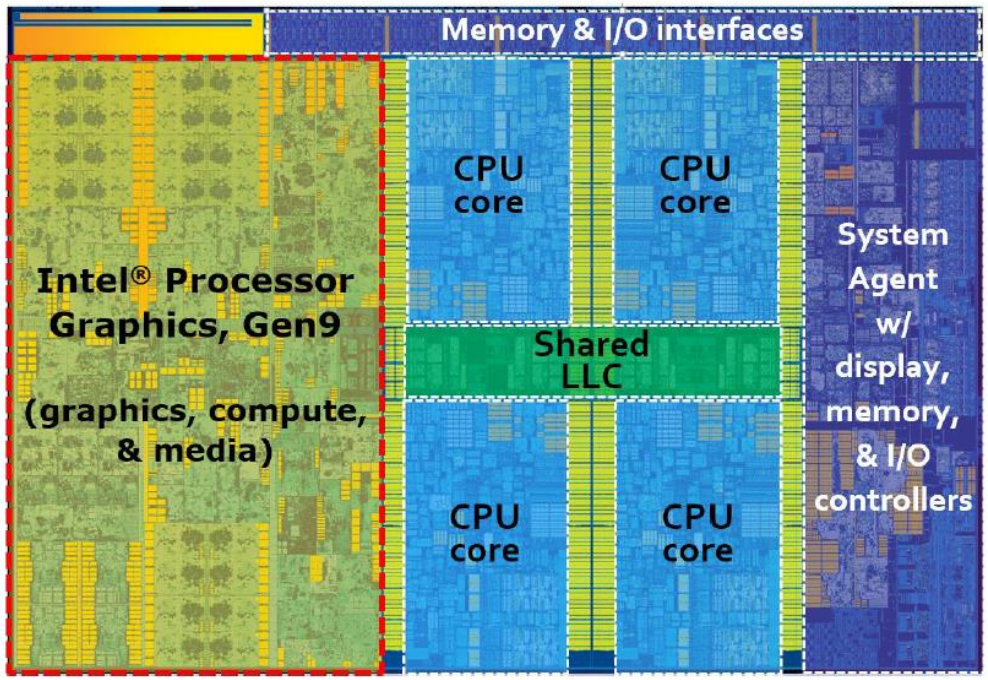


Figure 1: Architecture components layout for an Intel® Core™ i7 processor 6700K for desktop systems. This SoC contains 4 CPU cores, outlined in blue dashed boxes. Outlined in the red dashed box, is an Intel® HD Graphics 530. It is a one-slice instantiation of Intel processor graphics gen9 architecture.

x86 Clones: Advanced Micro Devices (AMD)

- **Historically**
  - AMD has followed just behind Intel
  - A little bit slower, a lot cheaper
- **Then**
  - Recruited top circuit designers from Digital Equipment Corp. and other downward trending companies
  - Built Opteron: tough competitor to Pentium 4
  - Developed x86-64, their own extension to 64 bits
- **Recent Years**
  - Intel got its act together
    - Leads the world in semiconductor technology
  - AMD has fallen behind
    - Relies on external semiconductor manufacturer

Intel's 64-Bit History

- **2001: Intel Attempts Radical Shift from IA32 to IA64**
  - Totally different architecture (Itanium)
  - Executes IA32 code only as legacy
  - Performance disappointing
- **2003: AMD Steps in with Evolutionary Solution**
  - x86-64 (now called "AMD64")
- **Intel Felt Obligated to Focus on IA64**
  - Hard to admit mistake or that AMD is better
- **2004: Intel Announces EM64T extension to IA32**
  - Extended Memory 64-bit Technology
  - Almost identical to x86-64!
- **All but low-end x86 processors support x86-64**
  - But, lots of code still runs in 32-bit mode

Our Coverage

- **IA32**
  - The traditional x86
  - For 15/18-213: RIP, Summer 2015
- **x86-64**
  - The standard
  - `shark> gcc hello.c`
  - `shark> gcc –m64 hello.c`
- **Presentation**
  - Book covers x86-64
  - Web aside on IA32
  - We will only cover x86-64

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- **Assembly Basics: Registers, operands, move**
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- C, assembly, machine code

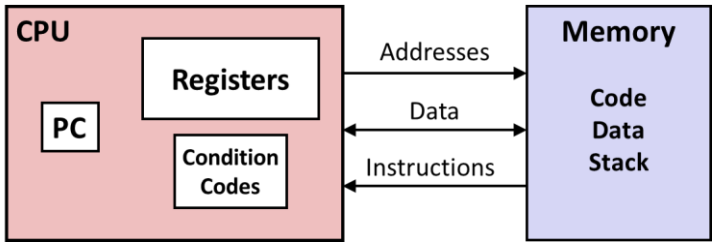
Levels of Abstraction

C programmer

C code

Nice clean layers, but beware…

Assembly programmer



Computer Designer

Caches, clock freq, layout, …

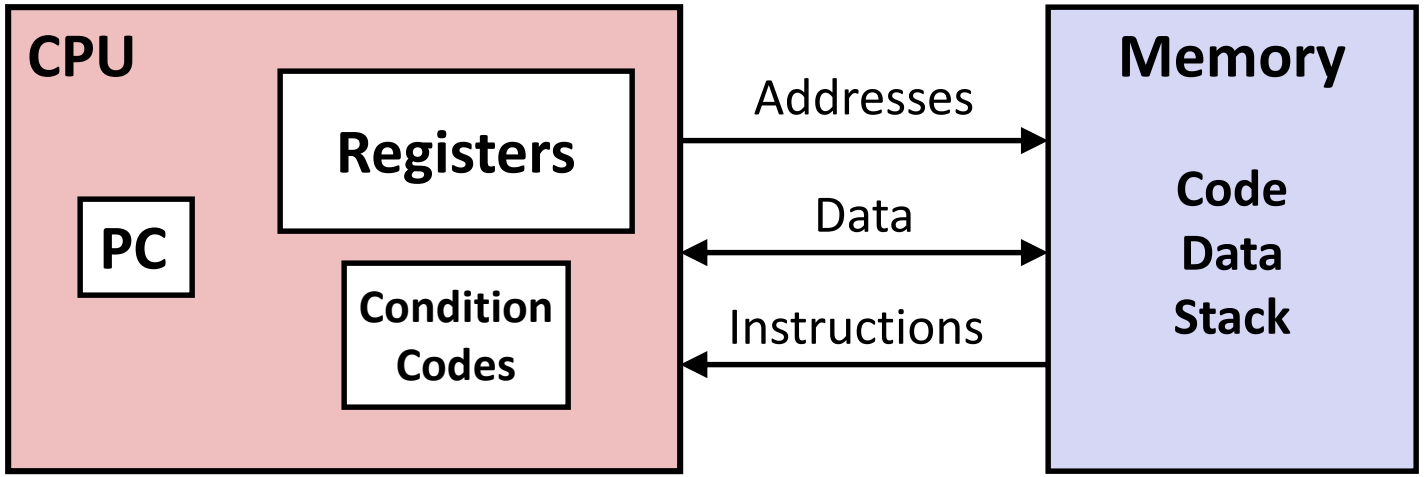
Of course, you know that: It's why you are taking this course.

Definitions

- **Architecture: (also ISA: instruction set architecture) The parts of a processor design that one needs to understand for writing assembly/machine code.**
  - Examples: instruction set specification, registers
- **Microarchitecture: Implementation of the architecture**
  - Examples: cache sizes and core frequency
- **Code Forms:**
  - Machine Code: The byte-level programs that a processor executes
  - Assembly Code: A text representation of machine code

- **Example ISAs:**
  - Intel: x86, IA32, Itanium, x86-64
  - ARM: Used in almost all mobile phones

Assembly/Machine Code View



Programmer-Visible State

- **PC: Program counter**
  - Address of next instruction
  - Called "RIP" (x86-64)
- **Register file**
  - Heavily used program data
- **Condition codes**
  - Store status information about most recent arithmetic or logical operation
  - Used for conditional branching

- **Memory**
  - Byte addressable array
  - Code and user data
  - Stack to support procedures

Assembly Characteristics: Data Types

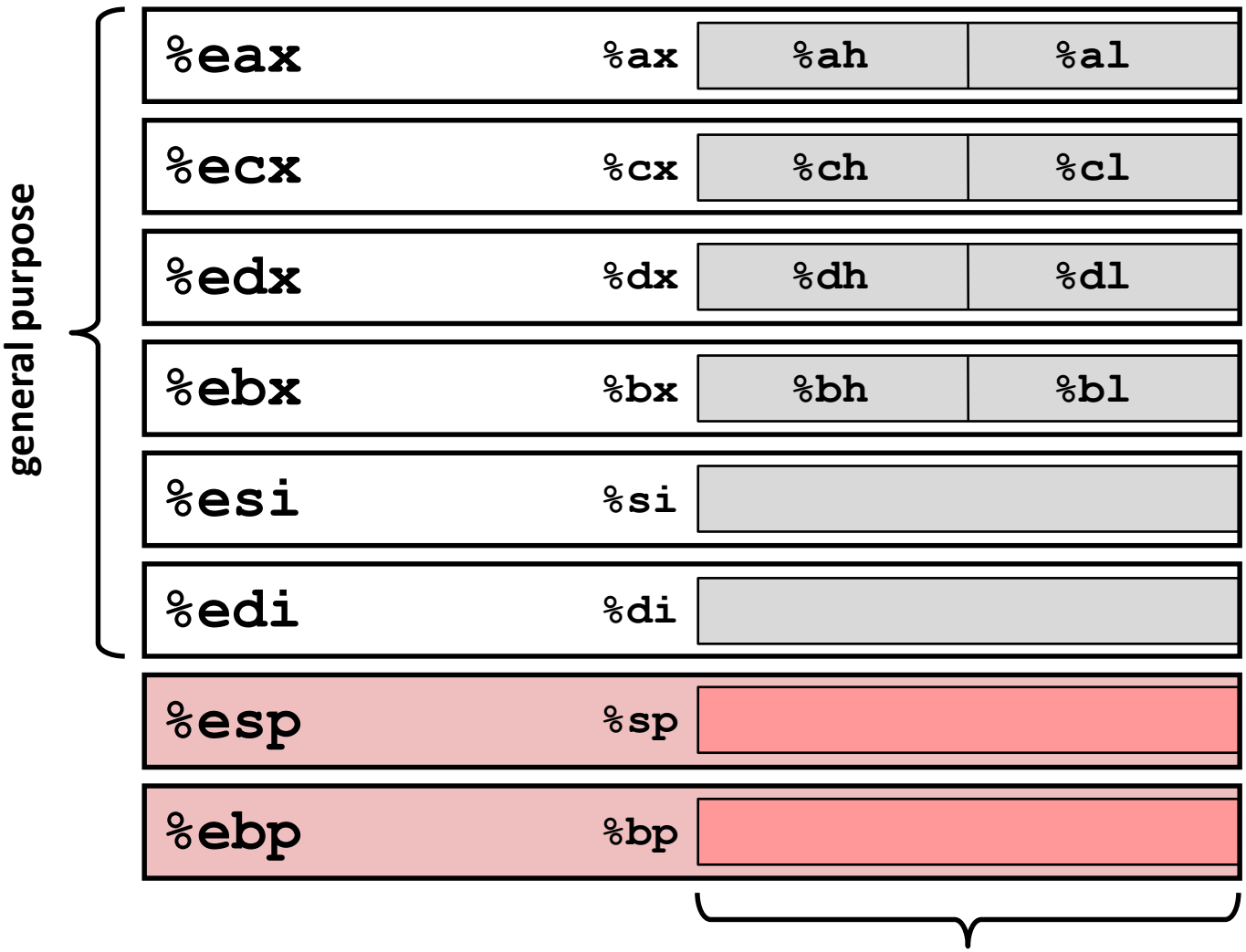
- **"Integer" data of 1, 2, 4, or 8 bytes**
  - Data values
  - Addresses (untyped pointers)
- **Floating point data of 4, 8, or 10 bytes**
- **Code: Byte sequences encoding series of instructions**
- **No aggregate types such as arrays or structures**
  - Just contiguously allocated bytes in memory

x86-64 Integer Registers

%rax	%eax	%r8	%r8d
%rbx	%ebx	%r9	%r9d
%rcx	%ecx	%r10	%r10d
%rdx	%edx	%r11	%r11d
%rsi	%esi	%r12	%r12d
%rdi	%edi	%r13	%r13d
%rsp	%esp	%r14	%r14d
%rbp	%ebp	%r15	%r15d

- Can reference low-order 4 bytes (also low-order 1 & 2 bytes)
- Not part of memory (or cache)

Some History: IA32 Registers



Assembly Characteristics: Operations

- **Transfer data between memory and register**
  - Load data from memory into register
  - Store register data into memory
- **Perform arithmetic function on register or memory data**
- **Transfer control**
  - Unconditional jumps to/from procedures
  - Conditional branches

Moving Data

- **Moving Data**

  `movq` ***Source, Dest***

- **Operand Types**
  - ***Immediate:*** Constant integer data
    - Example: `$0x400, $-533`
    - Like C constant, but prefixed with `'$'`
    - Encoded with 1, 2, or 4 bytes
  - ***Register:*** One of 16 integer registers
    - Example: `%rax, %r13`
    - But `%rsp` reserved for special use
    - Others have special uses for particular instructions
  - ***Memory:*** 8 consecutive bytes of memory at address given by register
    - Simplest example: `(%rax)`
    - Various other "addressing modes"

`%rax`
`%rcx`
`%rdx`
`%rbx`
`%rsi`
`%rdi`
`%rsp`
`%rbp`

`%rN`

Warning: Intel docs use mov *Dest, Source*

movq Operand Combinations

	Source	Dest	Src,Dest	C Analog
movq	*Imm*	*Reg*	`movq $0x4,%rax`	`temp = 0x4;`
		Mem	`movq $-147,(%rax)`	`*p = -147;`
	Reg	*Reg*	`movq %rax,%rdx`	`temp2 = temp1;`
		Mem	`movq %rax,(%rdx)`	`*p = temp;`
	Mem	*Reg*	`movq (%rax),%rdx`	`temp = *p;`

Cannot do memory-memory transfer with a single instruction

Simple Memory Addressing Modes

- **Normal (R) Mem[Reg[R]]**
  - Register R specifies memory address
  - Aha! Pointer dereferencing in C

```
movq (%rcx),%rax
```

- **Displacement D(R) Mem[Reg[R]+D]**
  - Register R specifies start of memory region
  - Constant displacement D specifies offset

```
movq 8(%rbp),%rdx
```

Example of Simple Addressing Modes

```
void
whatAmI(<type> a, <type> b)
{
      ????
}
```

%rdi

%rsi

```
whatAmI:
    movq    (%rdi), %rax
    movq    (%rsi), %rdx
    movq    %rdx, (%rdi)
    movq    %rax, (%rsi)
    ret
```

Example of Simple Addressing Modes

```
void swap
   (long *xp, long *yp)
{
  long t0 = *xp;
  long t1 = *yp;
  *xp = t1;
  *yp = t0;
}
```

```
swap:
   movq    (%rdi), %rax
   movq    (%rsi), %rdx
   movq    %rdx, (%rdi)
   movq    %rax, (%rsi)
   ret
```

Understanding Swap()

```
void swap
   (long *xp, long *yp)
{
  long t0 = *xp;
  long t1 = *yp;
  *xp = t1;
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}
```
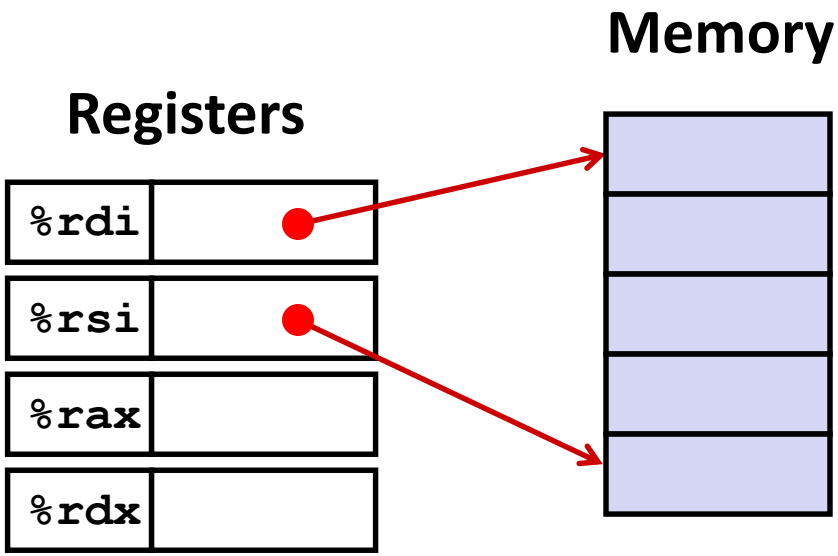



Register	Value
%rdi	xp
%rsi	yp
%rax	t0
%rdx	t1

```
swap:
   movq    (%rdi), %rax   # t0 = *xp
   movq    (%rsi), %rdx   # t1 = *yp
   movq    %rdx, (%rdi)   # *xp = t1
   movq    %rax, (%rsi)   # *yp = t0
   ret
```

Understanding Swap()

Registers

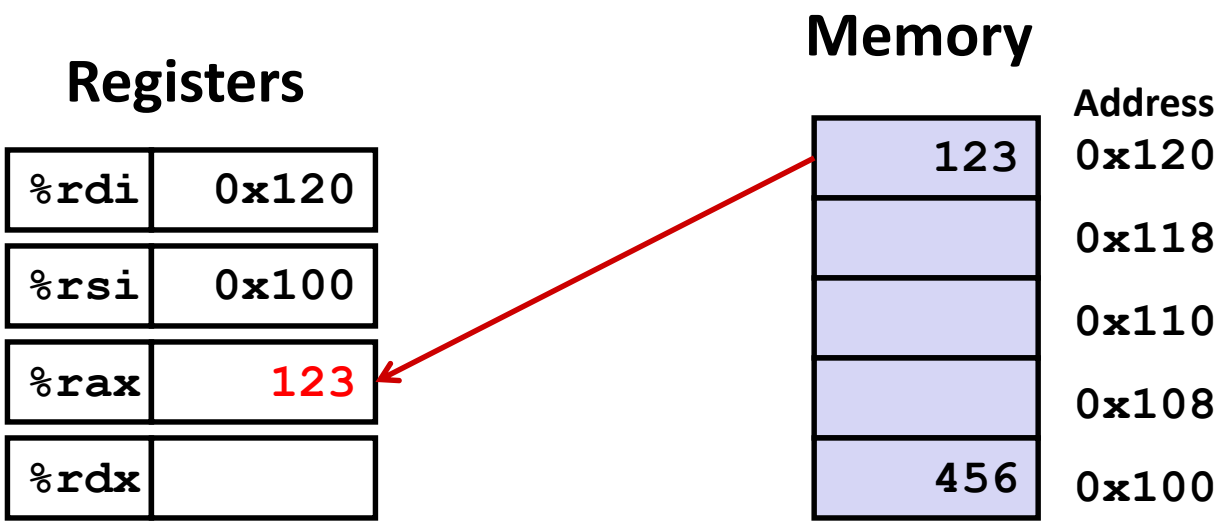
%rdi	0x120
%rsi	0x100
%rax	
%rdx	

Memory

	Address
123	0x120
	0x118
	0x110
	0x108
456	0x100

```
swap:
   movq    (%rdi), %rax   # t0 = *xp
   movq    (%rsi), %rdx   # t1 = *yp
   movq    %rdx, (%rdi)   # *xp = t1
   movq    %rax, (%rsi)   # *yp = t0
   ret
```

Understanding Swap()

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swap:
    movq    (%rdi), %rax   # t0 = *xp
    movq    (%rsi), %rdx   # t1 = *yp
    movq    %rdx, (%rdi)   # *xp = t1
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    ret
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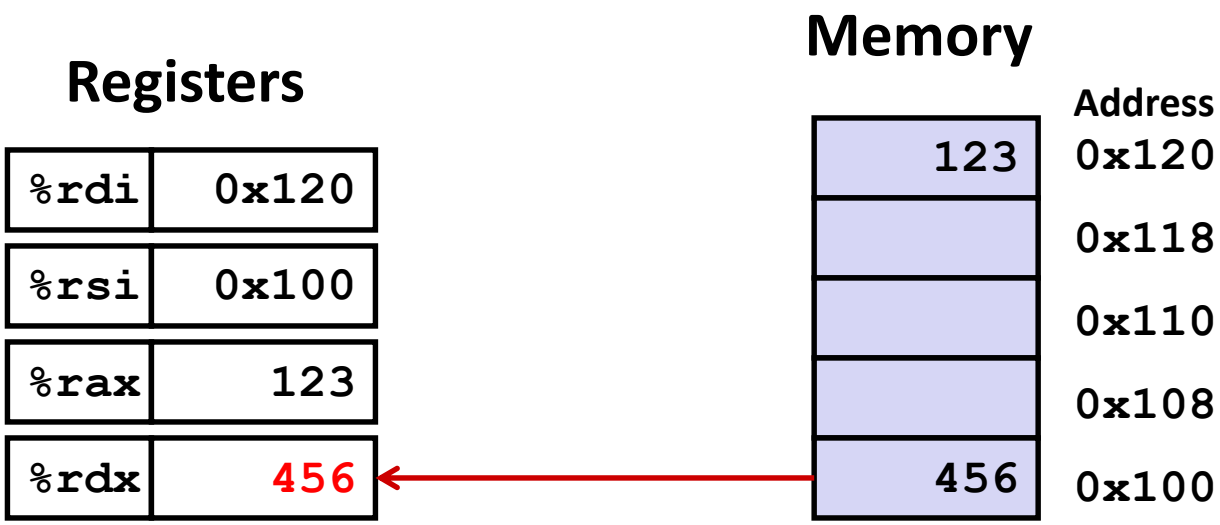
Understanding Swap()

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    movq    (%rdi), %rax   # t0 = *xp
    movq    (%rsi), %rdx   # t1 = *yp
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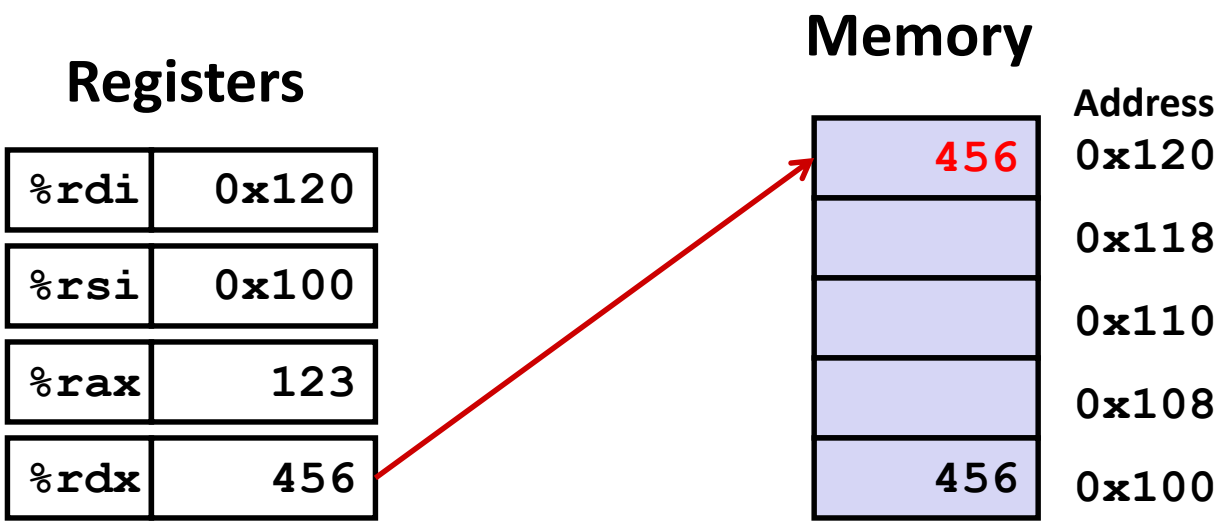
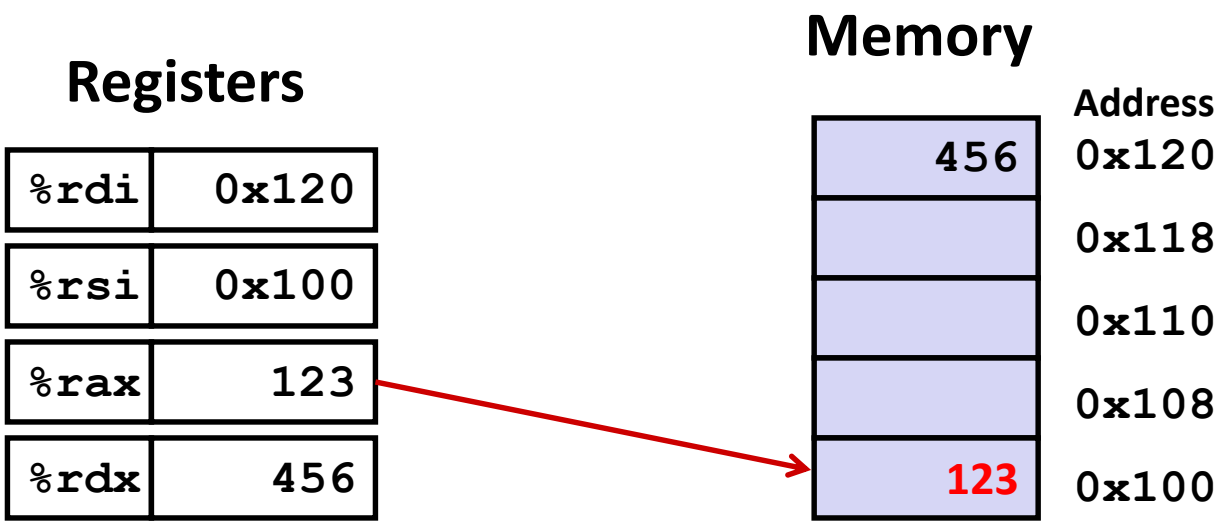



```
swap:
    movq    (%rdi), %rax   # t0 = *xp
    movq    (%rsi), %rdx   # t1 = *yp
    movq    %rdx, (%rdi)   # *xp = t1
    movq    %rax, (%rsi)   # *yp = t0
    ret
```

Understanding Swap()

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    movq    (%rdi), %rax   # t0 = *xp
    movq    (%rsi), %rdx   # t1 = *yp
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```

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  - Register R specifies start of memory region
  - Constant displacement D specifies offset

```
movq 8(%rbp),%rdx
```

Complete Memory Addressing Modes

- **Most General Form**

  D(Rb,Ri,S) **Mem[Reg[Rb]+S*Reg[Ri]+ D]**

  - D: Constant "displacement" 1, 2, or 4 bytes
  - Rb: Base register: Any of 16 integer registers
  - Ri: Index register: Any, except for `%rsp`
  - S: Scale: 1, 2, 4, or 8 (*why these numbers?*)

- **Special Cases**

  (Rb,Ri) **Mem[Reg[Rb]+Reg[Ri]]**

  D(Rb,Ri) **Mem[Reg[Rb]+Reg[Ri]+D]**

  (Rb,Ri,S) **Mem[Reg[Rb]+S*Reg[Ri]]**

Address Computation Examples

`%rdx`	`0xf000`
`%rcx`	`0x0100`

Expression	Address Computation	Address
`0x8(%rdx)`	`0xf000 + 0x8`	`0xf008`
`(%rdx,%rcx)`	`0xf000 + 0x100`	`0xf100`
`(%rdx,%rcx,4)`	`0xf000 + 4*0x100`	`0xf400`
`0x80(,%rdx,2)`	`2*0xf000 + 0x80`	`0x1e080`

Today: Machine Programming I: Basics

- History of Intel processors and architectures
- Assembly Basics: Registers, operands, move
- **Arithmetic & logical operations**
- C, assembly, machine code

Address Computation Instruction

- **`leaq` *Src, Dst***
  - *Src* is address mode expression
  - Set *Dst* to address denoted by expression

- **Uses**
  - Computing addresses without a memory reference
    - E.g., translation of `p = &x[i];`
  - Computing arithmetic expressions of the form x + k*y
    - k = 1, 2, 4, or 8

- **Example**

```
long m12(long x)
{
  return x*12;
}
```

Converted to ASM by compiler:

```
leaq (%rdi,%rdi,2), %rax   # t = x+2*x
salq $2, %rax              # return t<<2
```

Some Arithmetic Operations

- **Two Operand Instructions:**

Format	*Computation*		
`addq`	*Src,Dest*	Dest = Dest + Src	
`subq`	*Src,Dest*	Dest = Dest – Src	
`imulq`	*Src,Dest*	Dest = Dest * Src	
`salq`	*Src,Dest*	Dest = Dest << Src	***Also called shlq***
`sarq`	*Src,Dest*	Dest = Dest >> Src	***Arithmetic***
`shrq`	*Src,Dest*	Dest = Dest >> Src	***Logical***
`xorq`	*Src,Dest*	Dest = Dest ^ Src	
`andq`	*Src,Dest*	Dest = Dest & Src	
`orq`	*Src,Dest*	Dest = Dest | Src	

- **Watch out for argument order! *Src,Dest***
  (Warning: Intel docs use "op *Dest,Src*")
- **No distinction between signed and unsigned int (why?)**

Some Arithmetic Operations

- **One Operand Instructions**

`incq`	*Dest*	*Dest = Dest + 1*
`decq`	*Dest*	*Dest = Dest – 1*
`negq`	*Dest*	*Dest = – Dest*
`notq`	*Dest*	*Dest = ~Dest*

- **See book for more instructions**

Arithmetic Expression Example

```
long arith
(long x, long y, long z)
{
  long t1 = x+y;
  long t2 = z+t1;
  long t3 = x+4;
  long t4 = y * 48;
  long t5 = t3 + t4;
  long rval = t2 * t5;
  return rval;
}
```

```
arith:
  leaq    (%rdi,%rsi), %rax
  addq    %rdx, %rax
  leaq    (%rsi,%rsi,2), %rdx
  salq    $4, %rdx
  leaq    4(%rdi,%rdx), %rcx
  imulq   %rcx, %rax
  ret
```

Interesting Instructions

- **`leaq`**: address computation
- **`salq`**: shift
- **`imulq`**: multiplication
  - But, only used once

Understanding Arithmetic Expression Example

```
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(long x, long y, long z)
{
  long t1 = x+y;
  long t2 = z+t1;
  long t3 = x+4;
  long t4 = y * 48;
  long t5 = t3 + t4;
  long rval = t2 * t5;
  return rval;
}
```

```
arith:
  leaq    (%rdi,%rsi), %rax   # t1
  addq    %rdx, %rax          # t2
  leaq    (%rsi,%rsi,2), %rdx
  salq    $4, %rdx            # t4
  leaq    4(%rdi,%rdx), %rcx  # t5
  imulq   %rcx, %rax          # rval
  ret
```

Register	Use(s)
%rdi	Argument x
%rsi	Argument y
%rdx	Argument z, t4
%rax	t1, t2, rval
%rcx	t5

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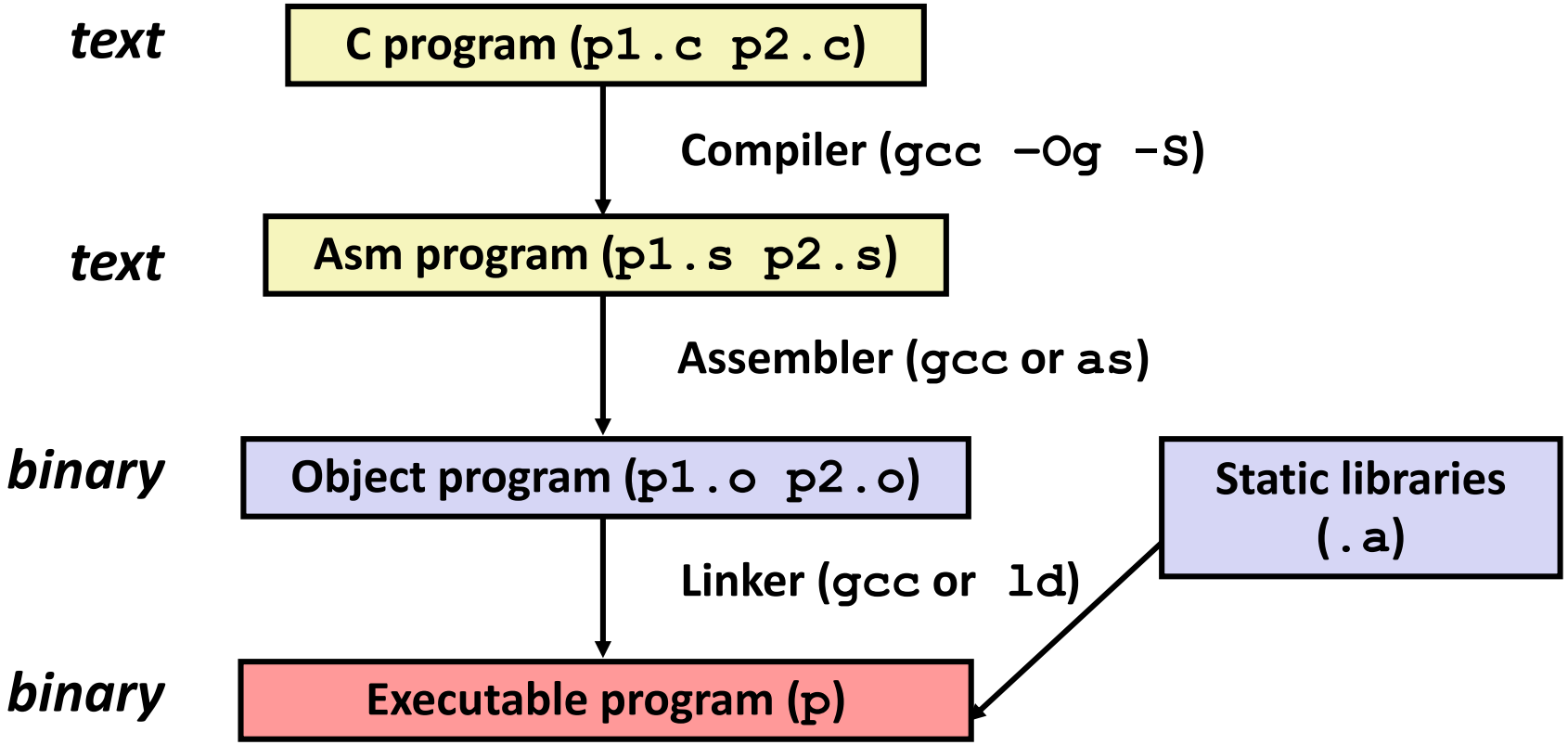
- **C, assembly, machine code**

Turning C into Object Code

- Code in files `p1.c p2.c`
- Compile with command: `gcc –Og p1.c p2.c -o p`
  - Use basic optimizations (`–Og`) [New to recent versions of GCC]
  - Put resulting binary in file `p`



Compiling Into Assembly

C Code (sum.c)

```
long plus(long x, long y);

void sumstore(long x, long y,
              long *dest)
{
    long t = plus(x, y);
    *dest = t;
}
```

Generated x86-64 Assembly

```
sumstore:
   pushq   %rbx
   movq    %rdx, %rbx
   call    plus
   movq    %rax, (%rbx)
   popq    %rbx
   ret
```

Obtain (on shark machine) with command

```
gcc –Og –S sum.c
```

Produces file `sum.s`

***Warning*: Will get very different results on non-Shark machines (Andrew Linux, Mac OS-X, …) due to different versions of gcc and different compiler settings.**

What it really looks like

```
        .globl  sumstore
        .type   sumstore, @function
sumstore:
.LFB35:
        .cfi_startproc
        pushq   %rbx
        .cfi_def_cfa_offset 16
        .cfi_offset 3, -16
        movq    %rdx, %rbx
        call    plus
        movq    %rax, (%rbx)
        popq    %rbx
        .cfi_def_cfa_offset 8
        ret
        .cfi_endproc
.LFE35:
        .size   sumstore, .-sumstore
```

What it really looks like

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        .globl  sumstore
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sumstore:
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        popq    %rbx
        .cfi_def_cfa_offset 8
        ret
        .cfi_endproc
.LFE35:
        .size   sumstore, .-sumstore
```

Things that look weird and are preceded by a '.' are generally directives.

```
sumstore:
   pushq   %rbx
   movq    %rdx, %rbx
   call    plus
   movq    %rax, (%rbx)
   popq    %rbx
   ret
```

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  - Store register data into memory
- **Perform arithmetic function on register or memory data**
- **Transfer control**
  - Unconditional jumps to/from procedures
  - Conditional branches

Object Code

Code for `sumstore`

```
0x0400595:
    0x53
    0x48
    0x89
    0xd3
    0xe8
    0xf2
    0xff
    0xff
    0xff
    0x48
    0x89
    0x03
    0x5b
    0xc3
```

- **Total of 14 bytes**
- **Each instruction 1, 3, or 5 bytes**
- **Starts at address `0x0400595`**

- **Assembler**
  - Translates `.s` into `.o`
  - Binary encoding of each instruction
  - Nearly-complete image of executable code
  - Missing linkages between code in different files
- **Linker**
  - Resolves references between files
  - Combines with static run-time libraries
    - E.g., code for **`malloc`**, **`printf`**
  - Some libraries are *dynamically linked*
    - Linking occurs when program begins execution

Machine Instruction Example

```
*dest = t;
```

```
movq %rax, (%rbx)
```

```
0x40059e:  48 89 03
```

- **C Code**
  - Store value **`t`** where designated by **`dest`**
- **Assembly**
  - Move 8-byte value to memory
    - Quad words in x86-64 parlance
  - Operands:

    `t`: Register **`%rax`**

    `dest`: Register **`%rbx`**

    `*dest`: Memory **M[`%rbx`]**
- **Object Code**
  - 3-byte instruction
  - Stored at address **`0x40059e`**

Disassembling Object Code

Disassembled

```
0000000000400595 <sumstore>:
  400595:  53                  push   %rbx
  400596:  48 89 d3            mov    %rdx,%rbx
  400599:  e8 f2 ff ff ff      callq  400590 <plus>
  40059e:  48 89 03            mov    %rax,(%rbx)
  4005a1:  5b                  pop    %rbx
  4005a2:  c3                  retq
```

- **Disassembler**

  `objdump -d sum`

  - Useful tool for examining object code
  - Analyzes bit pattern of series of instructions
  - Produces approximate rendition of assembly code
  - Can be run on either `a.out` (complete executable) or `.o` file

Alternate Disassembly

Disassembled

```
Dump of assembler code for function sumstore:
 0x0000000000400595 <+0>: push   %rbx
 0x0000000000400596 <+1>: mov    %rdx,%rbx
 0x0000000000400599 <+4>: callq  0x400590 <plus>
 0x000000000040059e <+9>: mov    %rax,(%rbx)
 0x00000000004005a1 <+12>:pop    %rbx
 0x00000000004005a2 <+13>:retq
```

- **Within gdb Debugger**
  - Disassemble procedure

```
gdb sum
disassemble sumstore
```

Alternate Disassembly

Object Code

```
0x0400595:
    0x53
    0x48
    0x89
    0xd3
    0xe8
    0xf2
    0xff
    0xff
    0xff
    0x48
    0x89
    0x03
    0x5b
    0xc3
```

Disassembled

```
Dump of assembler code for function sumstore:
 0x0000000000400595 <+0>: push   %rbx
 0x0000000000400596 <+1>: mov    %rdx,%rbx
 0x0000000000400599 <+4>: callq  0x400590 <plus>
 0x000000000040059e <+9>: mov    %rax,(%rbx)
 0x00000000004005a1 <+12>:pop    %rbx
 0x00000000004005a2 <+13>:retq
```

- **Within gdb Debugger**
  - Disassemble procedure

    `gdb sum`

    `disassemble sumstore`
  - Examine the 14 bytes starting at `sumstore`

    `x/14xb sumstore`

What Can be Disassembled?

```
% objdump -d WINWORD.EXE

WINWORD.EXE:   file format pei-i386

No symbols in "WINWORD.EXE".
Disassembly of section .text:

30001000 <.text>:
30001000:
30001001:
30001003:
30001005:
3000100a:
```

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- **Anything that can be interpreted as executable code**
- **Disassembler examines bytes and reconstructs assembly source**

Machine Programming I: Summary

- **History of Intel processors and architectures**
  - Evolutionary design leads to many quirks and artifacts
- **C, assembly, machine code**
  - New forms of visible state: program counter, registers, ...
  - Compiler must transform statements, expressions, procedures into low-level instruction sequences
- **Assembly Basics: Registers, operands, move**
  - The x86-64 move instructions cover wide range of data movement forms
- **Arithmetic**
  - C compiler will figure out different instruction combinations to carry out computation